ANALYSIS OF THE MONTHLY AIR TEMPERATURE (DEG. F) IN NOTTINGHAM CASTLE 1920-1939

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Introduction

In this project, we aim to analyse the time series titled "Mean Monthly Air Temperature (°F) at Nottingham Castle (1920–1939)". We begin by visualising the data to gain an initial understanding of its structure. This is followed by a series of preliminary transformations, including variance stabilisation, removal of seasonality, and elimination of any trend component.

After, we fit various seasonal ARIMA (S-ARIMA) models to the series, guided by the analysis of the ACF and PACF plots, as well as automatic model selection using the *Comp.Sarima()* function. The selected model is then evaluated through significance testing of the coefficients, analysis of the residuals, and comparison of prediction errors.

The project concludes with forecasting, using both the *predict()* function and the Holt-Winters exponential smoothing method.

1. Preliminary visualisation

The first step of the analysis is to visualise the time series to gain a better understanding of the data.

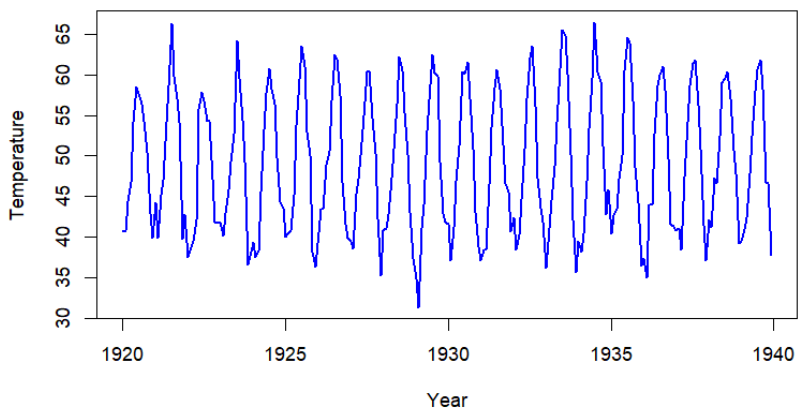

Figure 1: Mean monthly air temperature (°F) at Nottingham Castle (1920–1939)

The graph shows a strong seasonality, which is expected given that the data consists of monthly temperature records from 1920 to 1939 in Nottingham Castle. The time series clearly exhibits an annual cycle, suggesting a periodicity of $s = 12$.

The trend component, however, is difficult to observe from the graph. It suggests that the modeling of the time series may only require a seasonal component without a deterministic trend. However, before drawing any conclusions, it is necessary to perform a complementary analysis by overlaying the 12 months of each year to examine the seasonal behavior more closely.

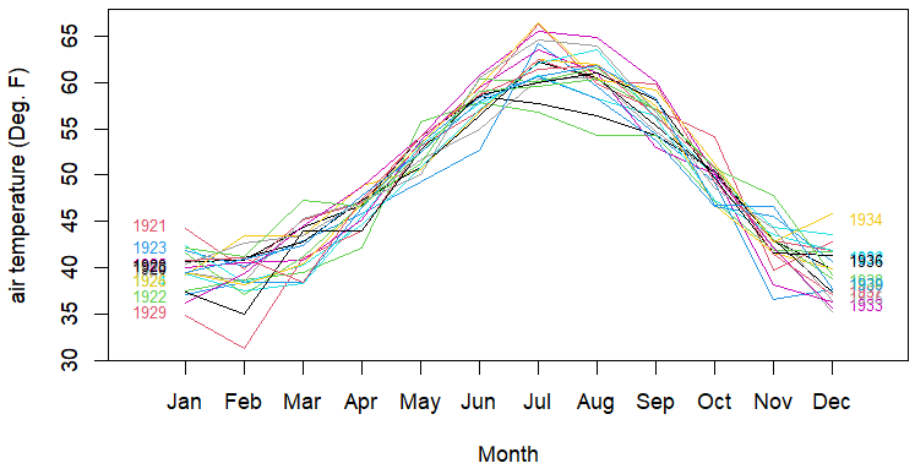

Figure 2: Seasonal patterns overlaid for each year.

This graph further supports the hypothesis of a negligible trend component. The seasonal patterns overlap closely from year to year, without a significant distinction between them. In time series with a trend, the seasonal curves tend to shift or diverge over time, which is not the case here.

2. Preliminary transformations

In this section, we are going to look at the variance, the trend and the seasonality of the time series. After that, we will be able to select a model in order to fit the time series.

2.1. Variance stabilisation

Before conducting a formal test for the trend, it is necessary to examine the variance of the time series. To address potential heteroskedasticity, a logarithmic transformation was applied.

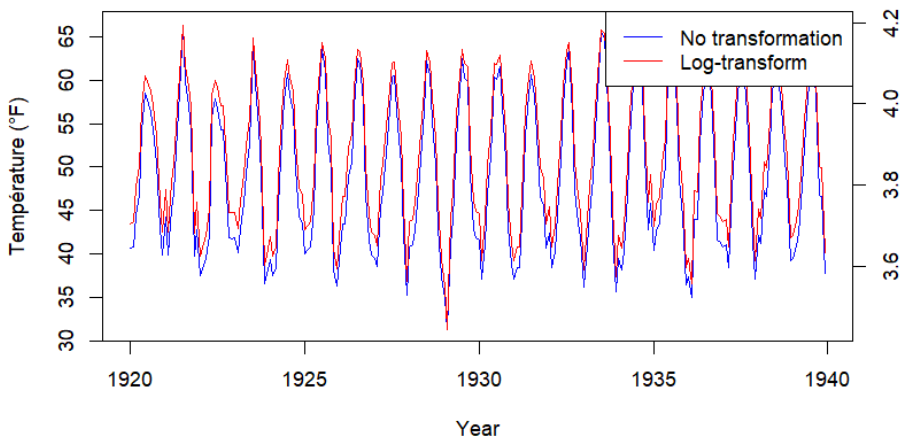

Figure 3: Comparison of original and log-transformed time series.

The log transformation was applied to the original time series. However, it did not significantly change the structure of the series. As we can see in the figure 3, the two curves almost entirely overlap, making it difficult to distinguish between them.

This observation suggests that the variance in the original time series is already relatively stable. Consequently, the log transformation is not necessary and will not be used in the subsequent analysis.

2.2. Suppresion of seasonality

The series shows a strong seasonal component, with a clear yearly cycle. To remove it, a seasonal differencing of period 12 was applied: $\nabla_{12} X_t = X_t - X_{t-12}$.

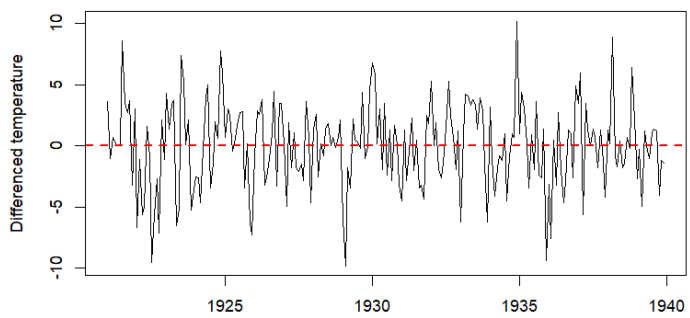

Figure 4: Time series without seasonality.

As we can see in figure 4, after this transformation, the times series looks to fluctuate around zero.

2.3. Suppresion of the trend

The final preliminary transformation considered is the removal of a potential trend component. This is typically performed using a first-order differencing ($d = 1$). However, rather than applying this transformation blindly, it is preferable to first test whether the time series is already stationary.

To this end, the Augmented Dickey-Fuller (ADF) test was conducted. The test result indicates a p-value below 5% (see **Appendix 1**), allowing us to reject the null hypothesis of a unit root. This suggests that the series is stationary in level, and therefore a differencing transformation is not required ($d = 0$).

This confirms that no persistent trend is present in the data, and the model can be built on the original series without differencing.

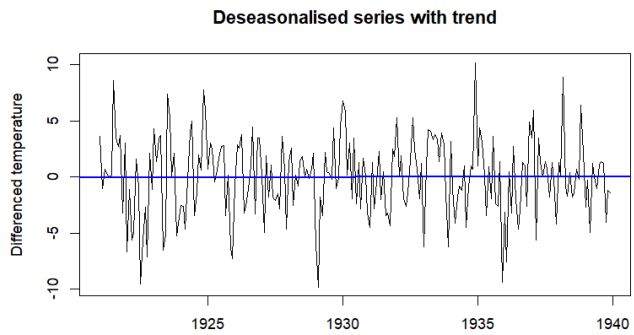

Figure 5: Deseasonalised series

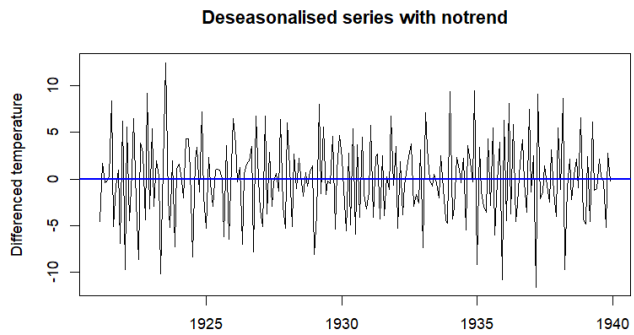

Figure 6: Deseasonalised series with differencing (lag 1)

The blue line represents the trend component of the series, which remains virtually unchanged regardless of whether the series is differenced.

3. Fitting of the S-ARIMA model

3.1. Interpretation of the ACF and PACF

We now look at the time series without seasonality, and her autocorrelation function (ACF) and her partial autocorrelation function (PACF).

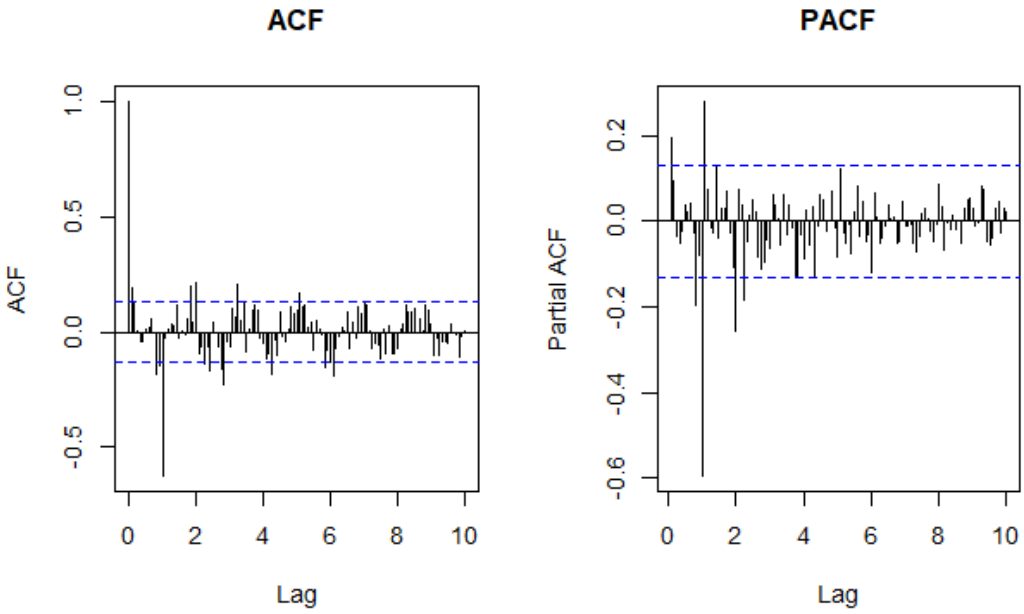

Figure 7: ACF and PACF (lags up to 120 months)

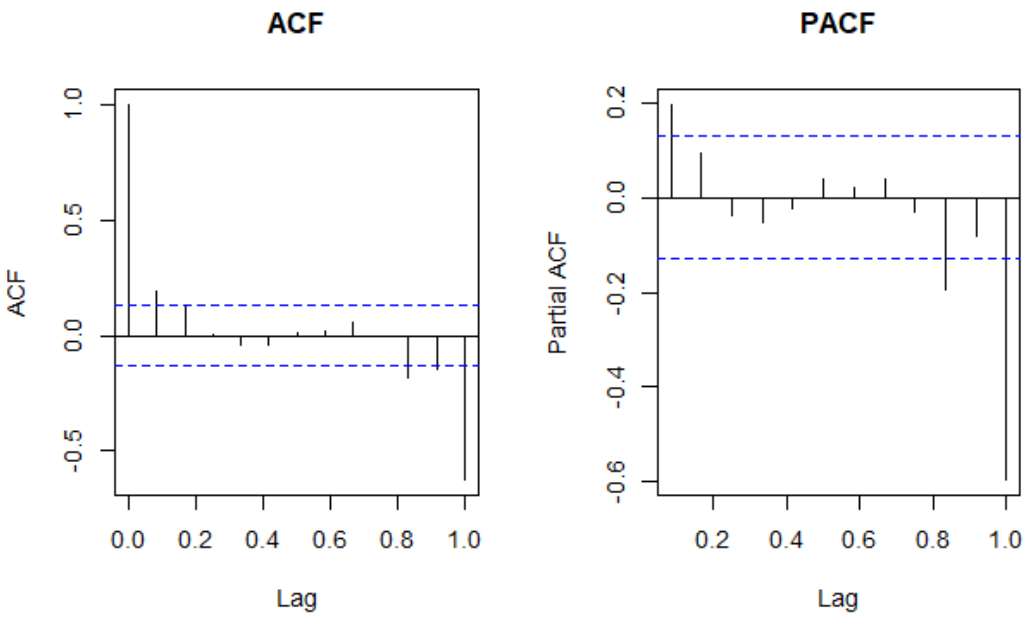

Figure 8: ACF and PACF (lags up to 12 months)

Based on the ACF and PACF plots, we observe one significant value at lag 1 in the ACF and two significant values at lags 1 and 2 in the PACF for the yearly dependence. This results suggests the possibility of an ARMA(1,2), ARMA(2,1), or ARMA(2,2) model.

For the monthly dependence, both the ACF and PACF show significant values only at lag 1, indicating that an AR(1), MA(1), or ARMA(1,1) model could be appropriate.

From this initial visual analysis, we can suggest that a SARIMA model of the form SARIMA$(p, 0, q) \times (P, 1, Q)_{12}$ may be suitable, with $p, q \leq 1$ and $P, Q \leq 2$.

Model	**AIC**
SARIMA$(1,0,0)(1,1,1)_{12}$	**1044.488**
SARIMA$(1,0,0)(1,1,2)_{12}$	1044.82
SARIMA$(1,0,0)(2,1,1)_{12}$	**1044.208**
SARIMA$(1,0,0)(2,1,2)_{12}$	1046.207
SARIMA$(1,0,1)(1,1,1)_{12}$	1045.546
SARIMA$(1,0,1)(1,1,2)_{12}$	1045.969
SARIMA$(1,0,1)(2,1,1)_{12}$	1045.305

Table 1: Comparison of SARIMA models based on AIC. The lowest AIC is highlighted in bold.

3.2. Selection of the models

Based on the AIC values presented in **Table 1**, we identify the two most promising models. The model with the lowest AIC is:

$$\text{SARIMA}(1,0,0) \times (2,1,1)_{12} \quad \text{with AIC} = \mathbf{1044.21}$$

The second-best model, with a slightly higher AIC, is:

$$\text{SARIMA}(1,0,0) \times (1,1,1)_{12} \quad \text{with AIC} = 1044.49$$

These two models will be examined in more detail.

3.3.Estimation of the two models

We know by definition that a time series $\{X_t\}$ is called SARIMA$(p, d, q) \times (P, D, Q)_s$ with period s, if:

$$Y_t = \nabla^d \nabla_s^D X_t = (1-B)^d (1-B^s)^D X_t$$

is a stationary ARMA process of the form:

$$A(B)F(B^s)Y_t = \Theta(B)G(B^s)\varepsilon_t$$

As said in the previous section, we consider the following two SARIMA models:
First model: SARIMA$(1,0,0) \times (2,1,1)_{12}$

$$(1-\phi_1 B)(1-\Phi_1 B^{12} - \Phi_2 B^{24})(1-B^{12})X_t = (1+\Theta_1 B^{12})\varepsilon_t$$

With the estimated coefficients:

$$\phi_1 = 0.2750, \quad \Phi_1 = -0.1923, \quad \Phi_2 = 0.1415, \quad \Theta_1 = -0.8157$$

Innovation variance: $\sigma^2 = 5.098$

Second model: SARIMA$(1,0,0) \times (1,1,1)_{12}$

$$(1-\phi_1 B)(1-\Phi_1 B^{12})(1-B^{12})X_t = (1+\Theta_1 B^{12})\varepsilon_t$$

With the estimated coefficients:

$$\phi_1 = 0.2739, \quad \Phi_1 = -0.2966, \quad \Theta_1 = -0.7306$$

Innovation variance: $\sigma^2 = 5.166$

3.4 Significance of the Coefficients

We now try to assess their statistical significance using the `coef.p()` function, as shown in **Appendix 3**. For the first model, SARIMA(1,0,0)(2,1,1)$_{12}$, two out of four coefficients were statistically significant at the 5% level. The second seasonal autoregressive coefficient (SAR(1)) and (SAR(2)) had a p-value above the threshold.

The second model, SARIMA(1,0,0)(1,1,1)$_{12}$, showed better results: all coefficients were highly significant, with p-values well below 0.01.

Based on these results, and considering both statistical significance and model parsimony, we choose to only proceed with the second model in the rest of the analysis.

3.5 Residuals diagnostic

We thus continue the analysis with the second model, and perform the figure 9.

Residual diagnostics looks good, the residuals are standardized and distributed around 0. We are not able to distinguish trend or seasolanity, the ACF of residuals indicates no significant autocorrelation. Finally, the Ljung-Box test fails to reject the null hypothesis of independence, meaning that the residuals behave like white noise.

Moreover, the autocorrelation function of the squared residuals does not show any pattern, suggesting the absence of conditional heteroskedasticity. The use of GARCH-type models is then not necessary in this case.

4. Prediction

The last section is the prediction of our fitted S-ARIMA to predict future time series observations. We will use the predict() function and Holt-Winters algorithm.

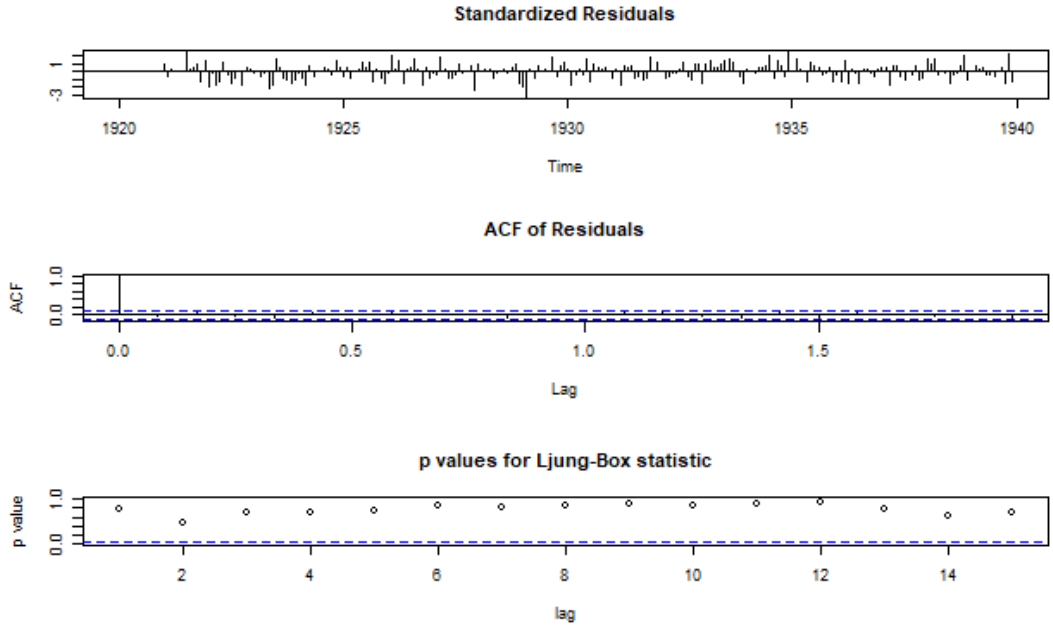

Figure 9: Deseasonalised series

4.1 Function predict()

Figures 10 and 11 present the predictions generated by the `predict()` function. The forecast is shown in orange, and the 95% confidence interval is represented by dashed lines.

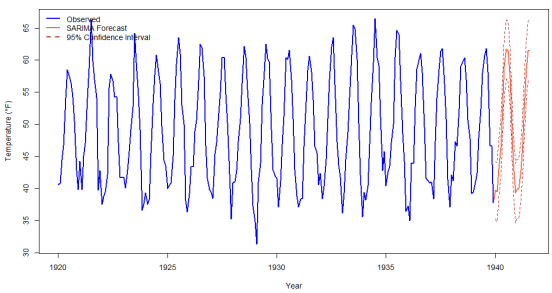

Figure 10: SARIMA prediction over the full time series

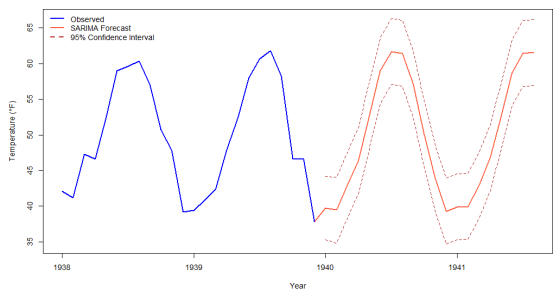

Figure 11: SARIMA prediction (zoom from 1938)

4.2 Holt-Winters algorithm

Figures 12 and 13 present similar before as the previous section. The forecast is shown in green, and the 95% confidence interval is represented by dashed lines.

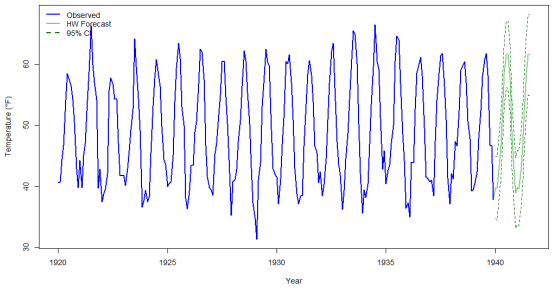

Figure 12: SARIMA prediction over the full time series

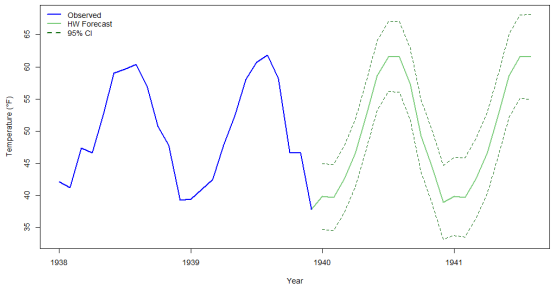

Figure 13: SARIMA prediction (zoom from 1938)

4.3 Comparaison of the two techniques

We observe that the confidence intervals for the Holt-Winters method are slightly wider, but both algorithms have very similar forecasts.

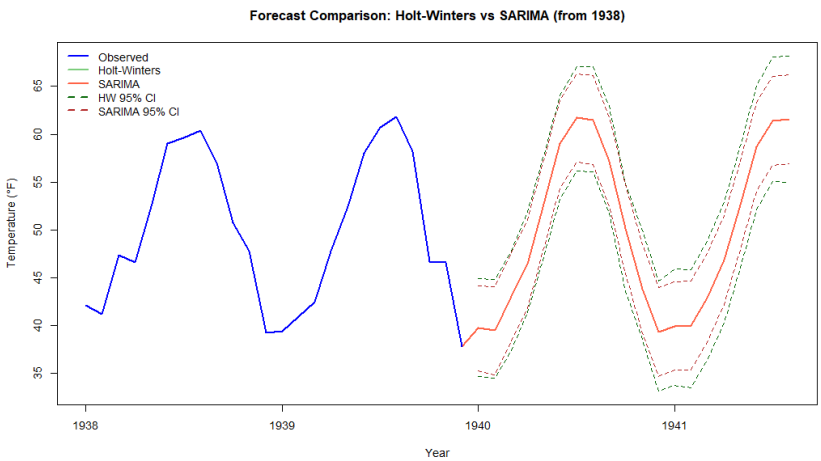

Figure 14: Comparaison of prediction with the predict() function and Holt-Winters algorithm

5. Conclusion

At the beginning of the analysis, we observed a clear seasonal pattern in the time series. Visual inspection raised some doubts about the presence of a trend, but further analysis suggested that no significant trend was present. Additionally, the variance appeared stable throughout the period.

The analysis of the ACF and PACF guided the selection of appropriate models. We initially compared the two models with the lowest AIC values. However, after identifying two non-significant coefficients in the first model, we decided to discard it and focus on the second model, which had all of its coefficients statistically significant.

We then examined the residuals of the selected model and applied two forecasting methods to evaluate its predictive performance.

Annexe

Appendix 1

Augmented Dickey-Fuller test (ADF):

Test statistic (Dickey-Fuller)	-5.3485
Lag order used	6
p-value	0.01
Alternative hypothesis	Stationary

Table 2: Results of the Augmented Dickey-Fuller test on the seasonally differenced series.

Appendix 2

Coefficient	**Estimate**	**Standard Error**
AR(1) (ϕ_1)	0.2750	0.0643
SAR(1) (Φ_1)	-0.1923	0.0995
SAR(2) (Φ_2)	0.1415	0.0932
SMA(1) (Θ_1)	-0.8157	0.0781
Variance (σ^2)	5.098	
Log-likelihood	-517.1	
AIC	1044.21	

Table 3: Estimated coefficients for the SARIMA(1,0,0)(2,1,1)$_{12}$ model.

Coefficient	**Estimate**	**Standard Error**
AR(1) (ϕ_1)	0.2739	0.0644
SAR(1) (Φ_1)	-0.2966	0.0817
SMA(1) (Θ_1)	-0.7306	0.0773
Variance (σ^2)	5.166	
Log-likelihood	-518.24	
AIC	1044.49	

Table 4: Estimated coefficients for the SARIMA(1,0,0)(1,1,1)$_{12}$ model.

Appendix 3

Model	**Coefficient**	**p-value**
SARIMA(1,0,0)(2,1,1)$_{12}$	AR(1)	1.90×10^{-5}
	SAR(1)	5.34×10^{-2}
	SAR(2)	1.29×10^{-1}
	SMA(1)	0.00
SARIMA(1,0,0)(1,1,1)$_{12}$	AR(1)	2.13×10^{-5}
	SAR(1)	2.81×10^{-4}
	SMA(1)	0.00

Table 5: p-values for the coefficients of the two SARIMA models. Values below 0.05 indicate significance at the 5% level.

R Code

```
# Load packages and data
library(tseries)
source("FonctionsSeriesChrono.r")
air <- read.table("air.txt", quote="\"", comment.char="")
air_temperature <- ts(air, start = c(1920, 1), frequency = 12)

# Plot original series
plot(air_temperature, xlab = "Year", ylab = "Temperature", col =
   "blue", lwd = 2)

# Seasonal pattern overlaid by year
years <- 1920:1939; nb_years <- length(years)
plot(air_temperature[1:12], type = "l", ylim = c(min(air_
   temperature), max(air_temperature)),
     xlim = c(0.2, 12.8), xaxt = "n", xlab = "Month", ylab = "air
         temperature (Deg. F)", col = 1)
for (i in 2:nb_years) {
  lines(air_temperature[12*(i-1)+(1:12)], col = i)
}
axis(1, at = 1:12, labels = month.abb)
for (i in 1:nb_years) {
  dy <- ifelse(i %% 2 == 0, 0.5, -0.5)
  if (i <= nb_years / 2) {
    x_pos <- 1
    y_pos <- air_temperature[12*(i-1) + 1] + dy
    text(x = x_pos - 0.3, y = y_pos, labels = years[i], col = i,
       cex = 0.7, adj = 1)
  } else {
    x_pos <- 12
    y_pos <- air_temperature[12*(i-1) + 12] + dy
    text(x = x_pos + 0.3, y = y_pos, labels = years[i], col = i,
       cex = 0.7, adj = 0)
  }
}

# Variance stabilisation
plot(air_temperature, xlab = "Year", ylab = "Temp rature ( F )",
    col = "blue", type = "l")
par(new = TRUE)
plot(log(air_temperature), axes = FALSE, xlab = "", ylab = "",
   col = "red", type = "l")
axis(side = 4); mtext("Log(Temp rature)", side = 4, line = 3)
legend("topright", legend = c("No transformation", "Log-transform
   "), col = c("blue", "red"), lty = 1)

# Deseasonalisation
air_temperature_noseason <- diff(air_temperature, lag = 12)
plot(air_temperature_noseason, main = "Deseasonalised series with
    trend", ylab = "Differenced temperature")
```

```
abline(h = 0, col = "red", lwd = 2, lty = 2)
trend_model <- lm(air_temperature_noseason ~ time(air_temperature
   _noseason))
abline(trend_model, col = "blue", lwd = 2)

# Remove trend
plot(air_temperature_noseason, main = "Deseasonalised series with
    notrend", ylab = "Differenced temperature")
trend_model <- lm(air_temperature_noseason_notrend ~ time(air_
   temperature_noseason_notrend))
abline(trend_model, col = "blue", lwd = 2)
adf.test(air_temperature_noseason)

# ACF and PACF
par(mfrow = c(1,2))
acf(air_temperature_noseason, main = "ACF", lag.max = 120)
pacf(air_temperature_noseason, main = "PACF", lag.max = 120)
par(mfrow = c(1,2))
acf(air_temperature_noseason, main = "ACF", lag.max = 12)
pacf(air_temperature_noseason, main = "PACF", lag.max = 12)

# Model fitting
Comp.Sarima(air_temperature, d=0, saison=12, D=1, p.max=1, q.max
   =1, P.max=2, Q.max=2, percentage_model=0.2)
modele1 <- arima(air_temperature, order = c(1,0,0), seasonal =
   list(order = c(2,1,1), period = 12))
modele2 <- arima(air_temperature, order = c(1,0,0), seasonal =
   list(order = c(1,1,1), period = 12))
coef.p <- function(estimates, std_errors) {
  pvals <- 2 * (1 - pnorm(abs(estimates / std_errors)))
  names(pvals) <- names(estimates)
  return(pvals)
}
coef.p(modele1$coef, sqrt(diag(modele1$var.coef)))
coef.p(modele2$coef, sqrt(diag(modele2$var.coef)))
tsdiag(modele2, gof.lag = floor(sqrt(length(air_temperature))))

# Holt-Winters Forecast
mod.HW <- HoltWinters(air_temperature)
pred.HW <- predict(mod.HW, n.ahead = 20, prediction.interval =
   TRUE)
ts.pred <- ts(pred.HW[, "fit"], start = end(air_temperature) + c
   (0,1), frequency = 12)
ts.upr <- ts(pred.HW[, "upr"], start = end(air_temperature) + c
   (0,1), frequency = 12)
ts.lwr <- ts(pred.HW[, "lwr"], start = end(air_temperature) + c
   (0,1), frequency = 12)
ts.full <- ts(c(air_temperature, ts.pred), start = start(air_
   temperature), frequency = frequency(air_temperature))
plot(ts.full, type = "l", col = "palegreen3", lwd = 2, ylab = "
   Temperature ( F )", xlab = "Year")
```

```
lines(air_temperature, col = "blue", lwd = 2)
lines(ts.upr, col = "darkgreen", lty = 2)
lines(ts.lwr, col = "darkgreen", lty = 2)
legend("topleft", legend = c("Observed", "HW Forecast", "95% CI")
   , col = c("blue", "palegreen3", "darkgreen"),
       lty = c(1, 1, 2), lwd = 2, bty = "n")

# SARIMA Forecast
pred.SARIMA <- predict(modele2, n.ahead = 20)
ts.pred.SARIMA <- ts(pred.SARIMA$pred, start = end(air_
   temperature) + c(0,1), frequency = 12)
ts.upr.SARIMA <- ts(pred.SARIMA$pred + 1.96 * pred.SARIMA$se,
   start = end(air_temperature) + c(0,1), frequency = 12)
ts.lwr.SARIMA <- ts(pred.SARIMA$pred - 1.96 * pred.SARIMA$se,
   start = end(air_temperature) + c(0,1), frequency = 12)
ts.full.SARIMA <- ts(c(air_temperature, ts.pred.SARIMA), start =
   start(air_temperature), frequency = frequency(air_temperature)
   )
plot(ts.full.SARIMA, type = "l", col = "tomato", lwd = 2, ylab =
   "Temperature ( F )", xlab = "Year")
lines(air_temperature, col = "blue", lwd = 2)
lines(ts.upr.SARIMA, col = "firebrick", lty = 2)
lines(ts.lwr.SARIMA, col = "firebrick", lty = 2)
legend("topleft", legend = c("Observed", "SARIMA Forecast", "95%
   Confidence Interval"),
       col = c("blue", "tomato", "firebrick"), lty = c(1, 1, 2),
          lwd = 2, bty = "n")

# Comparison Holt-Winters vs SARIMA
ts.real <- window(air_temperature, start = c(1938, 1))
full.SARIMA <- ts(c(ts.real, ts.pred.SARIMA), start = c(1938,1),
   frequency = 12)
full.HW <- ts(c(ts.real, ts.pred), start = c(1938,1), frequency =
    12)
plot(full.HW, col = "palegreen3", lwd = 2,
     main = "Forecast Comparison: Holt-Winters vs SARIMA (from
        1938)",
     xlab = "Year", ylab = "Temperature ( F )")
lines(full.SARIMA, col = "tomato", lwd = 2)
lines(ts.real, col = "blue", lwd = 2)
lines(ts.upr, col = "darkgreen", lty = 2)
lines(ts.lwr, col = "darkgreen", lty = 2)
lines(ts.upr.SARIMA, col = "firebrick", lty = 2)
lines(ts.lwr.SARIMA, col = "firebrick", lty = 2)
legend("topleft", legend = c("Observed", "Holt-Winters", "SARIMA"
   , "HW 95% CI", "SARIMA 95% CI"),
       col = c("blue", "palegreen3", "tomato", "darkgreen", "
          firebrick"),
       lty = c(1, 1, 1, 2, 2), lwd = 2, bty = "n")
```

Listing 1: R code used for time series analysis